



# ST BENEDICT'S

|                 |                                      |                  |                                |
|-----------------|--------------------------------------|------------------|--------------------------------|
| <b>SUBJECT</b>  | <b>Physical Science:<br/>Paper 1</b> | <b>DATE</b>      | 15 November 2021               |
| <b>GRADE</b>    | 11                                   | <b>MARKS</b>     | 200 Marks                      |
| <b>EXAMINER</b> | Mr D. du Toit                        | <b>MODERATOR</b> | Mrs S. Govender<br>Mr F. McCoy |
| <b>NAME</b>     |                                      | <b>DURATION</b>  | 3 hours                        |
| <b>CLASS</b>    |                                      |                  |                                |

**LOW ORDER**

50%

**HIGH ORDER**

50%

## INSTRUCTIONS

1. This question paper consists of 13 pages, an Answer Sheet of 2 pages and a Data Sheet of 2 pages. Please make sure that your question paper is complete.
2. Answer ALL the questions.
3. Read the questions carefully.
4. Use the data and formulae sheet whenever necessary.
5. Start each question on a new page.
6. **Show your working in all calculations.**
7. Units need not be included in the working of calculations, but appropriate units should be shown in all answers.
8. Express all answers to TWO decimal places unless told otherwise.
9. It is in your own interest to write legibly and to present your work neatly.

| Question | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | TOTAL |
|----------|----|----|----|----|----|----|----|----|----|-------|
| Total    | 20 | 25 | 28 | 17 | 20 | 24 | 27 | 26 | 13 | 200   |

## QUESTION 1

20 MARKS

Answer the following multiple-choice questions on your separate ANSWER SHEET. Make a cross (X) in the box corresponding to the letter which you consider to be the most correct.

- 1.1 Two displacement vectors of magnitudes 20 cm and 80 cm are added. Which one of the following is the ONLY possible choice for the magnitude of the resultant?

A 0 cm  
B 28 cm  
C 82 cm  
D 114 cm

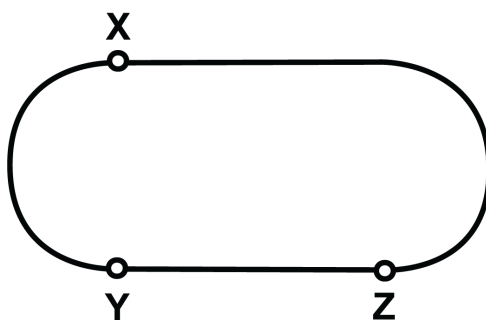
(2)

- 1.2 An object moves in the positive x-direction with positive velocity and a positive constant acceleration. After 3 s, its acceleration changes to a constant negative value. Which row of the table below is CORRECT concerning the velocity and displacement of the object immediately after 3 seconds?

|   | Velocity                       | Displacement |
|---|--------------------------------|--------------|
| A | Decreases and remains positive | Increases    |
| B | Decreases and becomes negative | Increases    |
| C | Increases and remains positive | Decreases    |
| D | Increases and becomes negative | Decreases    |

(2)

- 1.3 An athlete runs around the athletics track at a constant speed in an anticlockwise direction. The athletics track is shown in the diagram below.



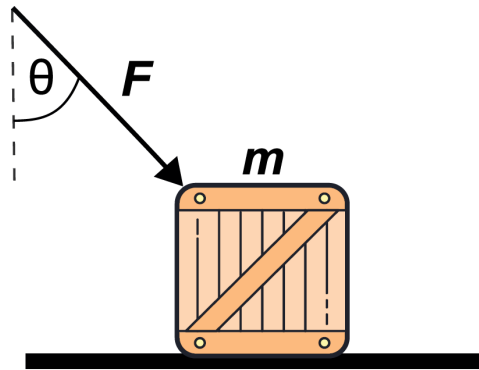
Which one of the following statements is true as the athlete runs past point X to Z via Y? The instantaneous velocity of the athlete is ...

A constant between X and Y only.  
B constant between Y and Z only.  
C constant at all points between X and Z.  
D never constant anywhere between X and Z.

(2)

- 1.4 An object thrown vertically upward into the air accelerates at  $9,8 \text{ m}\cdot\text{s}^{-2}$  downwards. Ignore the effects of air resistance. Which of the following statements is true?
- A The object travels  $9,8 \text{ m}$  during its first second.
  - B The object travels an extra  $9,8 \text{ m}$  during each second of its motion.
  - C The object speeds up at a rate of  $9,8 \text{ m}\cdot\text{s}^{-1}$  during each second of its motion.
  - D The speed of the object changes at a rate of  $9,8 \text{ m}\cdot\text{s}^{-1}$  during each second of its motion.
- (2)

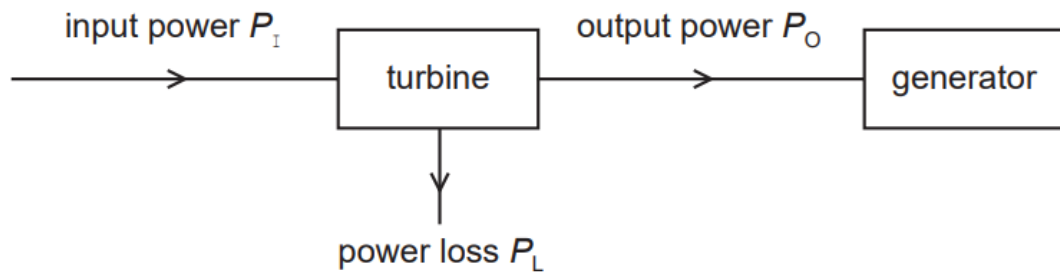
- 1.5 Consider the following diagram of a force  $F$  acting on a box of mass  $m$  as shown below. The box remains stationary.



Which of the following is the correct expression for the normal force acting on the box?

- A  $mg - F\cos\theta$
  - B  $mg + F\cos\theta$
  - C  $mg - F\sin\theta$
  - D  $mg + F\sin\theta$
- (2)
- 1.6 A person stands on a scale in an elevator. He notices that the scale reading is lower than his usual weight. Which of the following could possibly describe the motion of the elevator?
- A It is moving down at constant speed.
  - B It is moving down and slowing down.
  - C It is moving up and slowing down.
  - D It is moving up and speeding up.
- (2)
- 1.7 A laptop rests on a table. According to Newton's 3rd law, what is the reaction force to the weight of the laptop?
- A The upward force of the table on the laptop.
  - B The upward force of the laptop on the Earth.
  - C The downward force of the Earth on the laptop.
  - D The normal force on the laptop.
- (2)

- 1.8 A steam turbine is used to drive a generator. The input power to the turbine is  $P_I$  and the output power is  $P_O$ . The power loss in the turbine is  $P_L$ , as shown below.

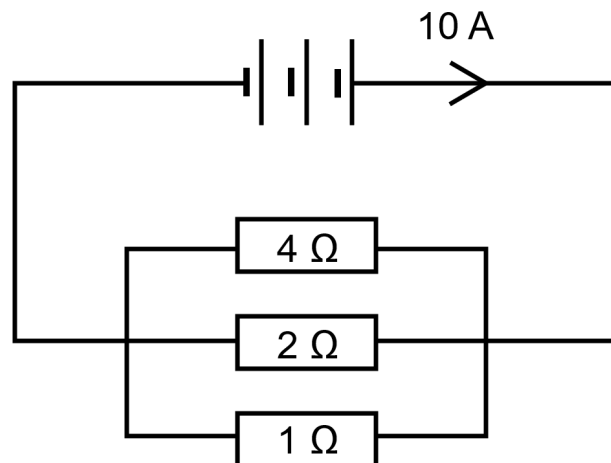


What is the efficiency of the turbine?

- A  $\frac{(P_I - P_L)}{P_I}$
- B  $\frac{P_I}{(P_I - P_L)}$
- C  $\frac{(P_I - P_L)}{P_O}$
- D  $\frac{P_O}{(P_I - P_L)}$

(2)

- 1.9 The current in a cell is 10 A has shown.

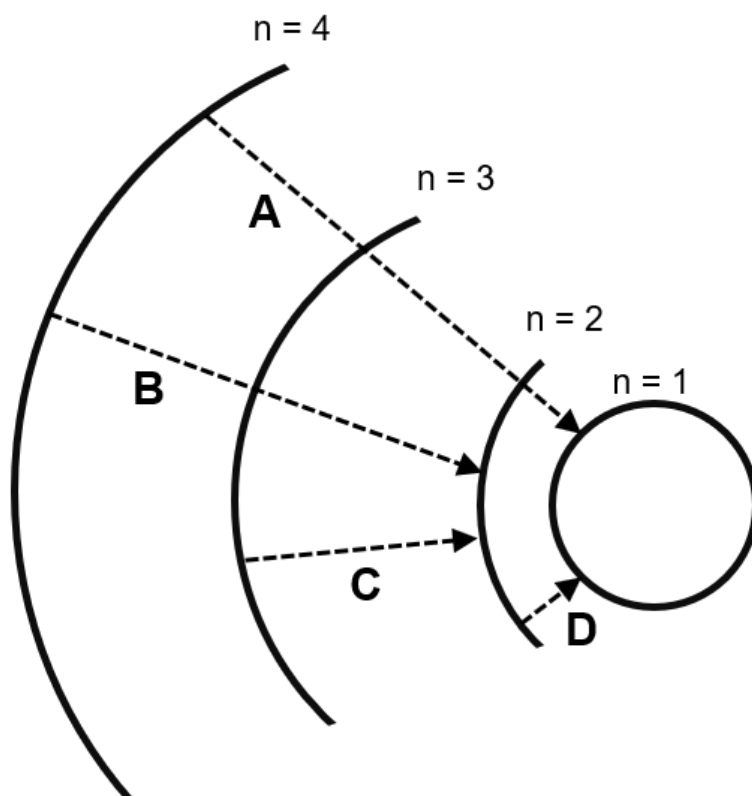


What is the current in the  $2\ \Omega$  resistor?

- A 0,35 A
- B 2,86 A
- C 3,50 A
- D 7,14 A

(2)

- 1.10 Consider the following energy level diagram for an atom of an element.  
Which electron transition corresponds to the light releases with the longest wavelength?

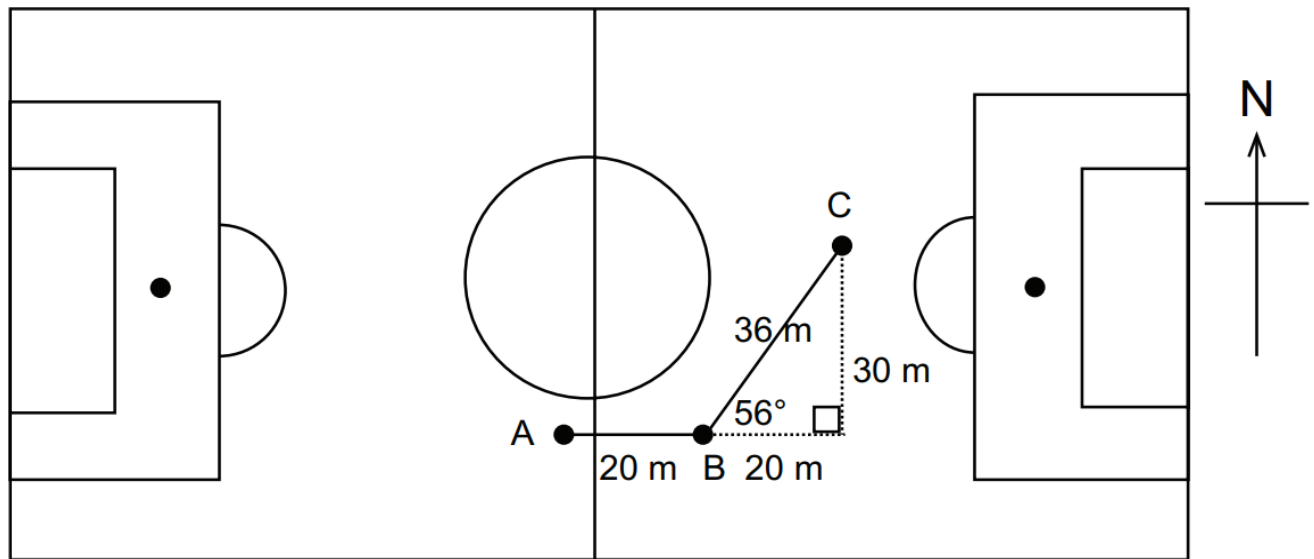


(2)

## QUESTION 2

25 MARKS

- 2.1 In a soccer match, a player runs 20 m due East in 10 s from position A to position B (as shown in the diagram below). He then breaks through the opposition's defence and is able to run to position C (on a bearing of  $34^\circ$ ) in another 10 s. Position C is 20 m East of B and 30 m North of B.

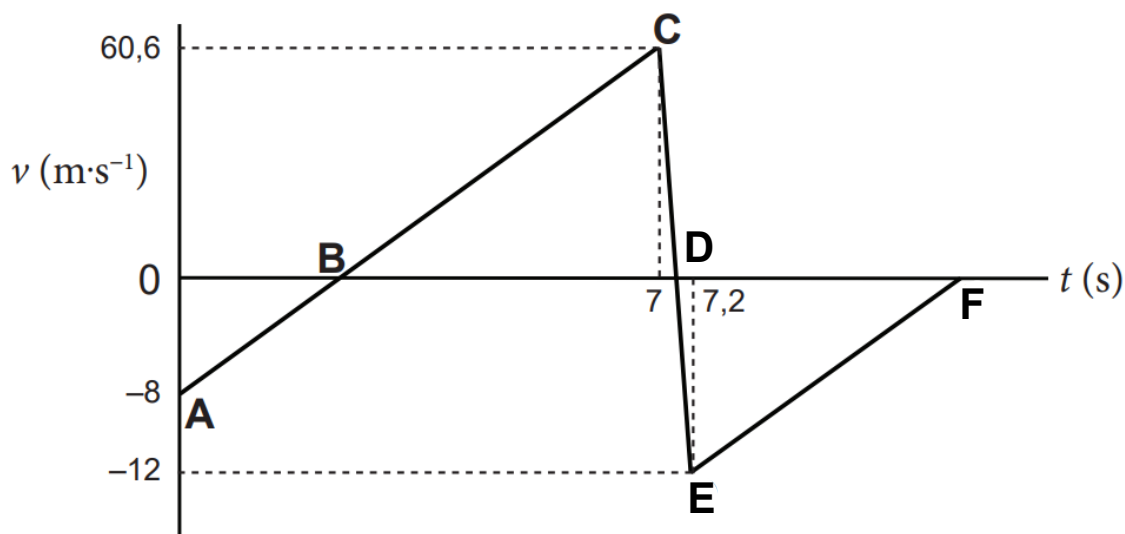


- 2.1.1 What is the difference between distance and displacement? Illustrate your answer with an example of each, referring to positions A, B and C, from the situation above. (4)
- 2.1.2 Define *speed*. (2)
- 2.1.3 Calculate the player's average speed from A to C. (3)
- 2.1.4 Draw a fully labelled  $90^\circ$  vector-addition triangle showing the player's resultant change in position from A to C. Include known values. (4)
- 2.1.5 Calculate magnitude of the player's average velocity from A to C. (5)
- 2.2 John waits on his bicycle at the end of his driveway for Paul to arrive. As Paul rides past on his bike with a constant speed of  $6 \text{ m.s}^{-1}$ , John gives him a 3 s head start before setting out after him. John starts from rest and rides with a constant acceleration of  $1,2 \text{ m.s}^{-2}$ . Assume rectilinear motion.
- 2.2.1 Calculate the time taken from when John sets off to when he catches up with Paul. (5)
- 2.2.2 Calculate the distance from John's driveway to the position where he catches up with Paul. (2)

**QUESTION 3****28 MARKS**

An object is projected vertically upwards from the top of a cliff. The object hits the ground at the base of the cliff and bounces upwards as shown in the diagram below.

The velocity versus time graph of the motion of the object is shown below:



- 3.1 Define *velocity*. (2)
- 3.2 What is the magnitude and direction of the initial velocity of the object? (2)
- 3.3 Describe the motion of the object from A to B. (2)
- 3.4 Calculate the time taken for the object to reach point B. (3)
- 3.5 Use the graph to calculate the distance the ball travelled from its maximum height to the ground. (3)
- 3.6 Use the graph to calculate the height of the cliff. (3)
- 3.7 Relative to the ground where is the object in its motion at time F? (2)
- 3.8 Draw a position versus time graph for the entire motion. Take the point of release as the reference point. Label your graph with the following values: (4)
- The value of the time taken for the object to reach its maximum height
  - The height of the cliff
  - Points A, B, C, D, E and F
- 3.9 Draw a free-body diagram showing all the forces acting on the ball at point D. (2)
- 3.10 If the mass of the ball is 0.6 kg calculate the magnitude of the force that the ground exerts on the ball while it is in contact with the ground. (5)

## QUESTION 4

17 MARKS

The photograph shows the result of an emergency braking carried out by a truck driver. He was driving a truck (mass  $10\,800\text{ kg}$ ) at  $22\text{ m}\cdot\text{s}^{-1}$  while carrying a granite block (mass  $160\,000\text{ kg}$ ) in the trailer. He applied brakes suddenly to avoid hitting another car

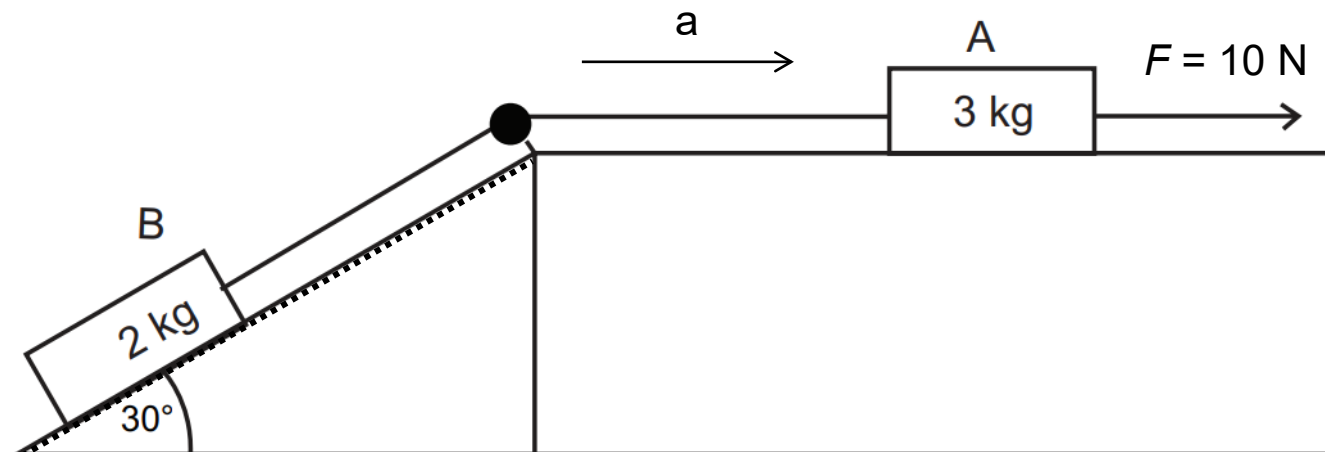


- 4.1 Draw a labelled free-body diagram showing the horizontal force(s) acting on the granite block while the truck is braking. You must also indicate the direction of the acceleration of the truck. (2)
  - 4.2 With reference to one of Newton's Laws, explain the movement of the granite block. (4)
  - 4.3 The tyre marks of the truck on the road were  $38,2\text{ m}$  in length from the time the driver slammed on the brakes to when the truck stopped.
    - 4.3.1 Define *acceleration*. (2)
    - 4.3.2 Calculate the acceleration of the truck. (4)
    - 4.3.3 Calculate the magnitude of the braking force exerted on the truck while it was brought to a standstill. (2)
    - 4.3.4 The driver states that the trucking company should place a rubber mat under such heavy loads. Making use of a relevant physics equation explain why this is a good idea to prevent such an accident in the future. (3)
-

## QUESTION 5

20 MARKS

Block A (3 kg) and block B (2 kg) are attached by means of a thin, inextensible cord. The cord is passed over a frictionless pulley. A horizontal force  $F$  of 10 N is applied to block A. Block A slides along a **smooth** horizontal surface and block B slides up a **rough** plane inclined at  $30^\circ$  to the horizontal. Block A experiences a frictional force of 14,7 N. The system accelerates as shown in the image below.

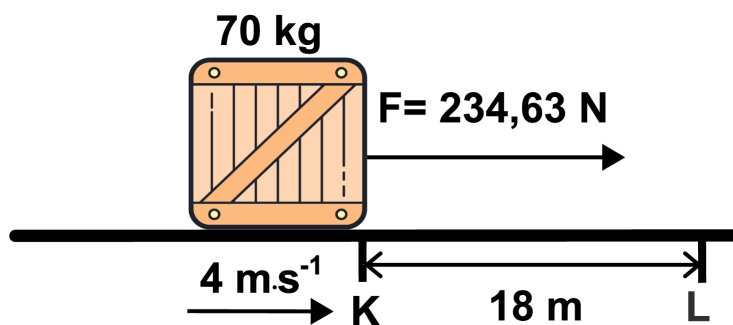


- 5.1 Draw a fully labelled free-body diagram showing all the force(s) acting on block B. (4)
- 5.2 Define *friction force*. (2)
- 5.3 Calculate the magnitude of the frictional force acting on block B if the coefficient of kinetic friction is 0,5. (3)
- 5.4 State *Newton's second law of motion*. (2)
- 5.5 Calculate the acceleration of the system (5)
- 5.6 Calculate the magnitude of the tension in the cord. (2)
- 5.7 While block B is accelerating up the slope, the cord snaps. How does the acceleration of block A change after the cord snaps? Write INCREASE, DECREASE or REMAIN THE SAME. (2)

## QUESTION 6

24 MARKS

- 6.1 A smart cart is pushed up a slope and allowed to free-wheel up the track until it comes to rest at the top of the track without hitting the bumper. The cart then moves back down the ramp and the student stops the cart at the bottom of the track.
- 6.1.1 Define *work*. (2)
- 6.1.2 Explain what is meant by “positive” and “negative” work? (2)
- 6.1.3 Which force always does negative work when the cart moves up and down the slope? (1)
- 6.1.4 How does the acceleration of the cart up the slope compare the acceleration of the cart down the slope. Explain your answer using a relevant physics equation. (3)
- 6.1.5 Sketch a velocity versus time graph for the motion of the cart up and down the track. Take down the ramp as negative. Label the point where the cart comes to rest as X. (3)
- 6.2 A crate of mass 70 kg slides along a rough, horizontal surface with a constant applied force of 234,63 N. The crate experiences a constant frictional force of 190 N.
- The crate passes point K at a speed of  $4 \text{ m.s}^{-1}$  before passing point L 18 m further along.

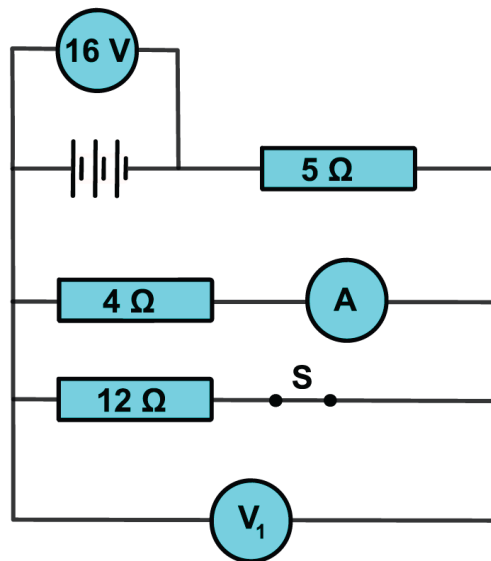


- 6.2.1 Which forces, on the crate, do no work. Explain briefly. (3)
- 6.2.2 Calculate the kinetic energy of the crate at point K. (3)
- 6.2.3 State the *work-energy theorem*. (2)
- 6.2.4 Use the work-energy theorem to calculate the speed of the crate as it passes point L. (5)

## QUESTION 7

27 MARKS

- 7.1 Study the circuit diagram shown below. The battery has an emf of 16 V. The ammeter and cells has negligible resistance and no current flows through the voltmeter. Switch S is closed.

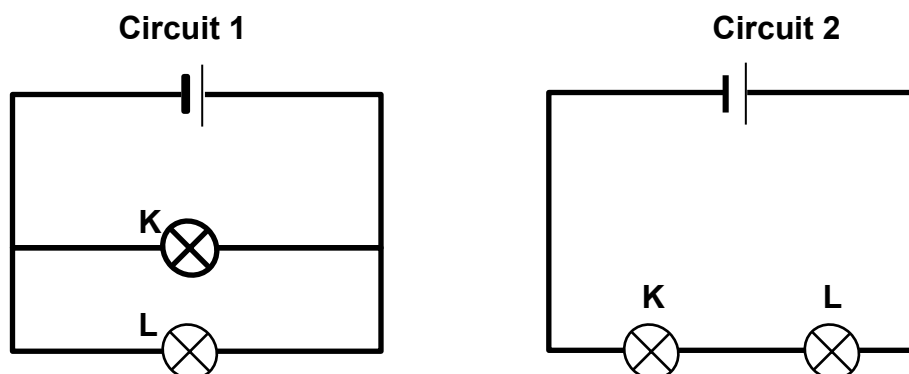


- 7.1.1 Define *resistance*. (2)
- 7.1.2 Calculate the effective resistance of the circuit. (4)
- The 4 Ω resistor has a power rating of 9 W. Calculate ...
- 7.1.3 the reading on ammeter A, (4)
- 7.1.4 the reading on voltmeter  $V_1$ , (3)
- 7.1.5 the current through the 5 Ω resistor. (3)

Switch S is now OPENED.

- 7.1.6 What will happen to the reading on the voltmeter connected across the battery? State either INCREASES, DECREASES or REMAINS THE SAME. (1)
- 7.1.7 What will happen to the reading on voltmeter  $V_1$ . State either INCREASES, DECREASES or REMAINS THE SAME. Fully explain your making use of a relevant physics formula. (4)

- 7.2 The same cell with no internal resistance is connected in turn to each of the two circuits shown below. Bulb K has twice the resistance of Bulb L.



- 7.2.1 Define *power*. (2)
- 7.2.2 How does the brightness of Bulb K compare with that of Bulb L in each of these circuits? Explain your answer. (4)

## QUESTION 8

26 MARKS

A simulation is used to prove Newton's law of gravitation. Represented by the equations below:

$$F = \frac{Gm_1m_2}{r^2}$$

Where:

F = the Gravitational force between the objects (N)

$m_1, m_2$  = the masses of the two objects (kg)

r = the distance between the object's centres (m)

G = Constant of Universal Gravitation. ( $6,7 \times 10^{-11} \text{N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$ )

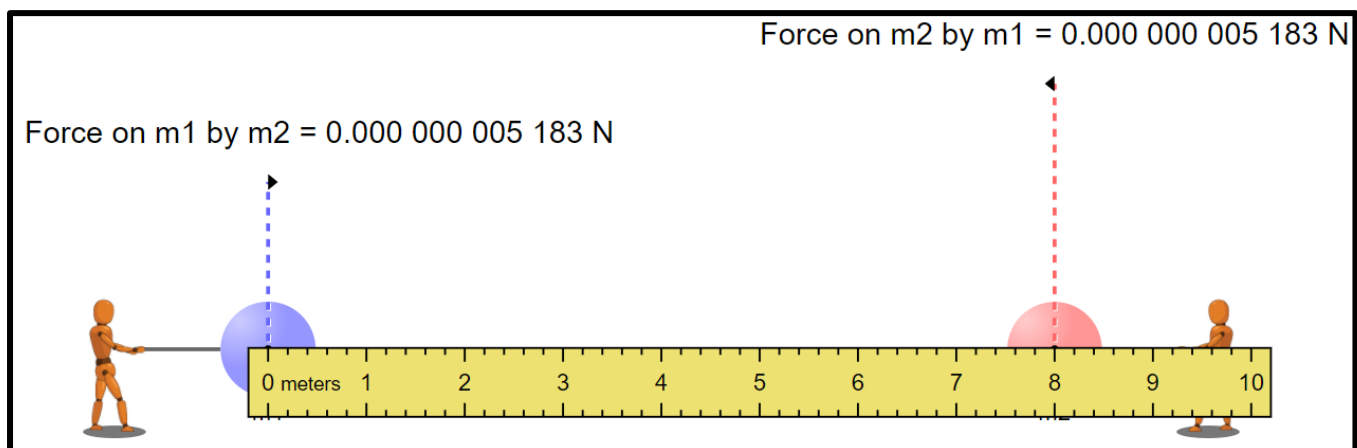
The table below shows the gravitational force of attraction between 2 identical spheres when placed different distances apart.

| Distance between centres (m) | Force experienced ( $\times 10^{-8} \text{ N}$ ) | Inverse of distance between centres squared ( $\text{m}^{-2}$ ) |
|------------------------------|--------------------------------------------------|-----------------------------------------------------------------|
| 4                            | 2,073                                            | 0,063                                                           |
| 5                            | 1,327                                            | 0,040                                                           |
| 6                            | 0,921                                            | 0,027                                                           |
| 7                            | 0,677                                            | 0,020                                                           |
| 8                            |                                                  |                                                                 |

8.1 The diagram below shows the masses at 8 meters apart.

8.1.1 Read off the force experienced and write it in the table on the ANSWER SHEET. (2)

8.1.2 Calculate the value in the last column write it in the table on the ANSWER SHEET. (2)

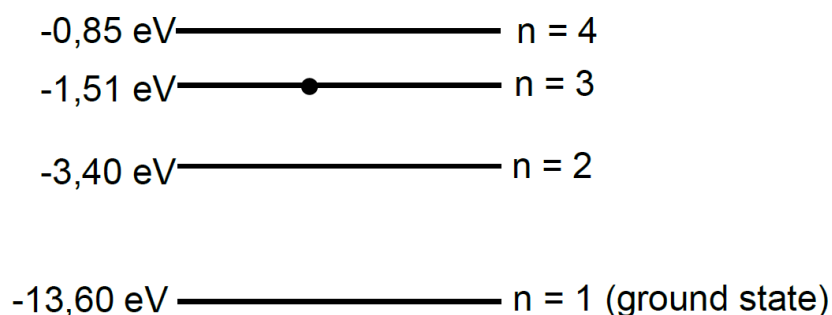


- 8.2 Plot a graph of inverse of distance squared vs force on the ANSWER SHEET. YOU MUST USE THE SCALE THAT HAS BEEN GIVEN. (7)
- 8.3 Write the relationship between these variables in words. (2)
- 8.4 Use your graph to determine the force these spheres would experience if they were 4,5 m apart from each other. Show how you determined this on your graph. (3)
- 8.5 Calculate the gradient of the graph. Include a unit. (4)
- 8.6 What physical quantities are represented by the gradient? (3)
- 8.7 Hence, use your answer in 8.5 and 8.6 to determine the mass of each spheres. (3)
- 

## QUESTION 9

13 MARKS

The energy levels for a hydrogen atom are represented on the diagram below.



An excited electron is in the 3rd energy level as indicated.

- 9.1 How many unique frequencies will be in the emission spectrum as the excited electron returns to the ground state? (1)
- 9.2 Calculate the frequency of the photon emitted when the electron transition is from the  $n=2$  level to the ground state (4)
- Hydrogen gas in a discharge tube emits light of wavelength 655 nm.
- 9.3 Calculate the energy (in eV) that corresponds to this wavelength of light. (4)
- 9.4 Which transition that will result in a wavelength of 655 nm? (2)
- 9.5 Why are the energy levels stated as negative values? (2)
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**END OF EXAMINATION**